

15. Admin and Ops Specification

Detailed operating model, information architecture, screen requirements, and workflows

Project: QuantPipeline

Document status: Working baseline for product and engineering delivery

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Audience: Product owner, architecture, development, QA, and future implementation agents

Purpose: This document is intended to be implementable. It defines contracts, boundaries, and decisions that should survive future feature growth without breaking the core recommendation pipeline.

Document Control

Field	Value
Document code	15
Title	Admin and Ops Specification
Primary input set	Documents 01-09, A1 technology recommendation, and baseline truth table
Primary architecture anchor	Data & Feature Layer -> Signals -> Selection -> Allocation -> Timing -> Risk Overlay -> Recommendation Object -> Backtest / Paper / Replay / Ops
Assumed release posture	Private single-user first release with future-safe modular boundaries
Key dependencies	Documents 10-14, 06 UX Appendix, 08 PRD, 09 Functional Requirements
Quality rule	No silent degradation, explicit trust semantics, and replayable output

1. Purpose and Scope

This document specifies the operator-facing side of QuantPipeline: the admin and ops workspace, its information architecture, module definitions, screen behavior, workflows, permissions, and acceptance logic.

The admin workspace is not a back-office afterthought. It is a product-quality layer. It should explain why the system is healthy or unhealthy, allow governed actions on publication and policy state, and preserve a clean audit trail for every meaningful decision.

2. Admin Personas

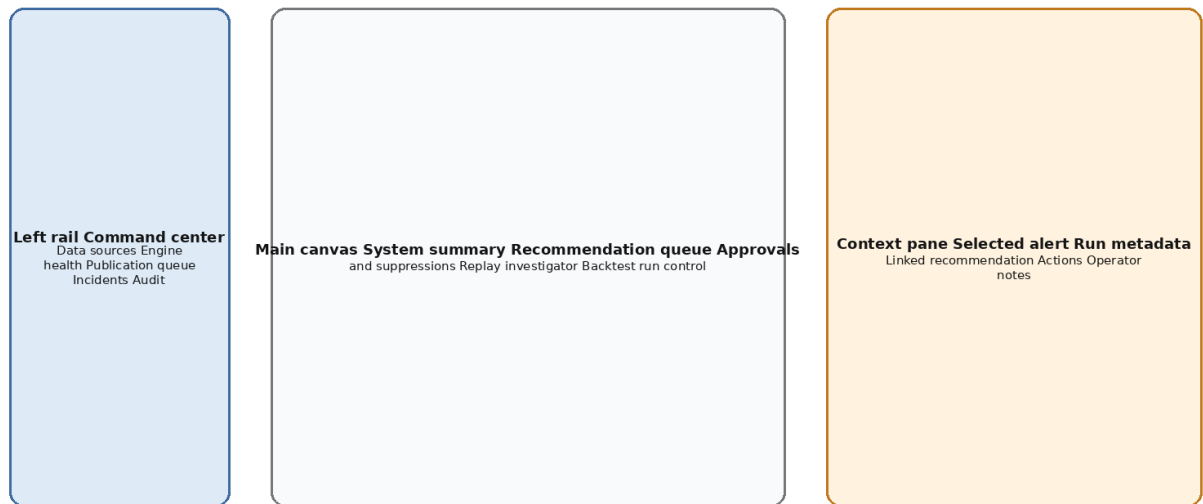
The first release may have one real person acting in multiple capacities, but the workspace should still recognize distinct modes of work.

Persona	Primary questions	Typical actions
Operator	Is the pipeline healthy? What failed? What needs attention?	Review alerts, inspect runs, annotate incidents
Approver	Is this recommendation safe to publish?	Review staged publication, approve or suppress
Administrator	Are policies, permissions, and guardrails correctly configured?	Promote bundles, manage controls
Forensic reviewer	Why did this recommendation happen?	Open replay, compare runs, inspect evidence history

3. Admin Information Architecture

Admin and Ops Workspace

Document 15 reference layout



Interaction pattern: queue -> inspect -> decide -> annotate -> preserve audit trail.

Figure 15-1. Admin and ops workspace layout.

Admin / Ops Workspace

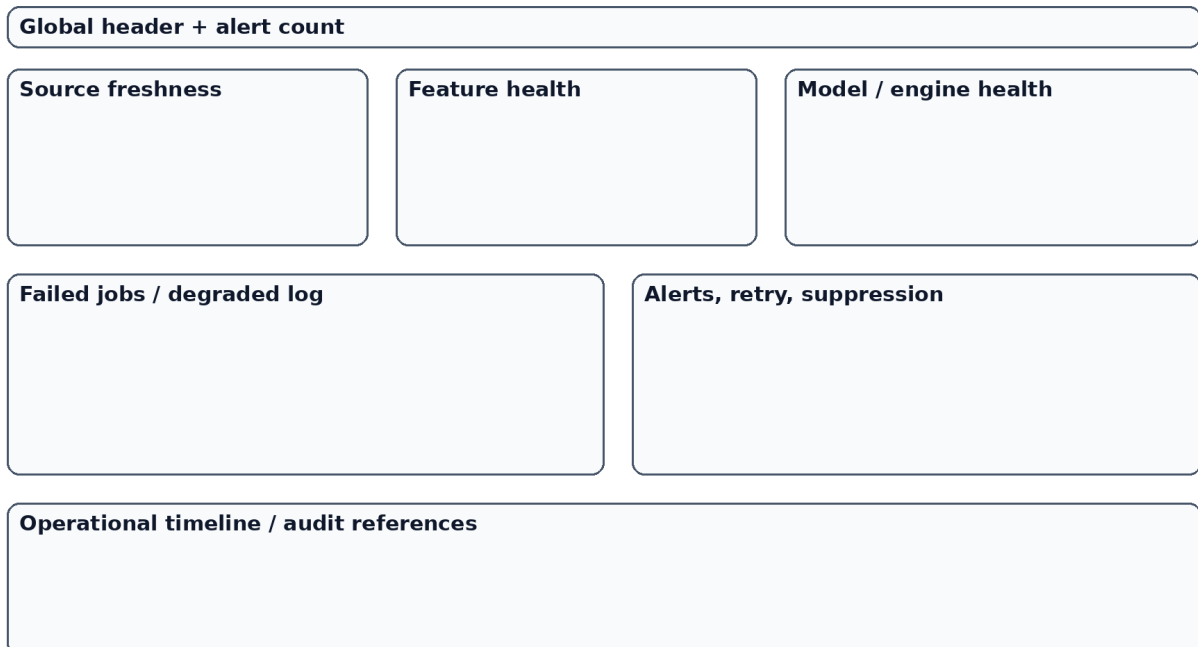


Figure 15-2. Prior admin concept retained as reference.

The recommended layout is a three-part workspace: a left rail for navigation and queue switching, a main canvas for the active module, and a right context pane for selected object details and actions. This keeps operational navigation stable while allowing deep inspection without losing queue context.

4. Core Workspace Rules

- The command center is the default landing screen for admins and operators.
- Every queue item should expose severity, freshness, linked object, and current owner if assigned.
- High-risk actions must require confirmation and capture a decision note.
- Operator notes should be visible in context but clearly distinguished from system facts.
- The right pane should preserve context while navigating between linked objects such as alert -> incident -> recommendation -> replay.
- No module should require raw log access for basic understanding of state.
- All actions that change publication, policy, or incident state must emit audit events.

5. Module Inventory

The admin and ops workspace should be organized into modules that map to distinct operator questions.

Module	Purpose	Primary widgets	Key actions
Command Center	Overall health and pending work	Status cards, queue summary, freshness panel	Open queue item, assign owner
Data Source Control	Source and freshness oversight	Feed status table, source history, exceptions	Acknowledge issue, open incident
Engine Health	Signal and policy execution health	Run timelines, failure matrix, engine versions	Re-run, inspect artifact, suppress if needed
Publication Queue	Approve or suppress staged recommendations	Recommendation list, diff summary, warning badges	Approve, suppress, request rerun
Replay Investigator	Historical forensic review	Replay timeline, stage diffs, evidence links	Pin finding, export note
Backtest Lab Ops	Validation job management	Experiment queue, scorecards, artifact list	Start job, compare runs, mark reviewed
Paper Monitor	Paper drift and shadow performance	Portfolio summary, rebalance log, benchmark chart	Trigger refresh, annotate discrepancy
Policies and Guardrails	Manage bundle versions and controls	Policy catalog, active bundle card, threshold table	Promote, rollback, edit threshold
Incidents and Audit	Investigate and prove actions	Incident board, audit explorer, notes	Create incident, resolve, export evidence

Module	Purpose	Primary widgets	Key actions
Settings and Access	Permissions and environment controls	Role matrix, environment summary	Update roles, rotate access

6. Command Center Specification

The command center should answer four questions immediately: Is the system healthy, what recommendation is current, what is blocked, and what requires a human decision now.

Recommended components: global health banner, recommendation freshness card, active incidents widget, publication queue widget, top alerts list, and validation backlog summary.

Area	Required content
Global banner	overall health state, active severity, current recommendation status
Freshness card	age of current recommendation inputs, age of latest publication
Pending decisions	count of staged recommendations and approvals waiting
Incidents	active incidents by severity with direct links
Backlog	stuck jobs, failed runs, pending validation reviews

7. Publication Queue Specification

This module is the operational heart of governed recommendation release. Each row should represent one recommendation pending a decision or recently affected by a publication event.

Each queue row should include recommendation id, as-of time, proposed state, overall confidence, freshness state, top warnings, changed weights summary, linked incident if any, and expected decision deadline where relevant.

Queue action	Behavior	Audit requirement
Approve	Publish or mark current recommendation active	Decision note required if warnings exist
Publish with warning	Allow publication in degraded state	Mandatory warning note
Suppress	Block recommendation from being current guidance	Reason code and linked incident required
Request rerun	Send recommendation back for regeneration	Trace job request
Assign reviewer	Transfer decision ownership	Ownership change event

8. Data Source and Engine Health Modules

These modules should make technical failures legible in decision terms. A stale source should show which recommendation contexts are affected. A failed engine should show whether the current recommendation is blocked, degraded, or unaffected.

Widget	Displays	Why it matters
Source status table	source, freshness age, last success, affected contexts	Connects feed health to user impact
Run timeline	last N runs by stage with success/failure	Shows whether problems are isolated or systemic
Failure matrix	stage by environment or by engine	Quick diagnosis of failure clusters
Version panel	promoted engine and policy versions	Helps explain unexpected output changes

9. Replay Investigator Specification

The replay investigator should support a forensic workflow: choose a historical recommendation, inspect chronology, compare stage outputs, identify what changed, and annotate findings.

The screen should preserve a stable left-to-right flow: context summary, chronological narrative, stage diff tabs, evidence references, and operator notes.

10. Backtest Lab Ops and Paper Monitor

Backtest Lab Ops is where validation jobs are started, compared, and reviewed. It should not masquerade as a generic data science notebook. It is an operational lab linked to promotion governance.

Paper Monitor is where the operator sees whether the production pipeline is behaving plausibly under live conditions. It should highlight divergence between expected rebalances and observed paper drift, benchmark-relative movement, and unresolved execution assumptions.

Module	Critical screens
Backtest Lab Ops	Experiment queue, run detail, comparison view, scorecard, promotion readiness view
Paper Monitor	Paper summary, holdings, drift analysis, rebalance history, discrepancy notes

11. Policies and Guardrails Module

This module manages the active policy bundle and guardrail thresholds that shape recommendation publication. It should show not only the active version, but also what changed versus the previous version, which evidence package supported the promotion, and what rollback option exists.

Editing thresholds directly in production should be rare. The module should favor versioned changes that can be reviewed and rolled back.

12. Incidents and Audit Module

The incident board should support severity, status, owner, linked object references, time opened, and current mitigation summary. The audit explorer should support filters by object, actor, action type, time, and severity.

Export of audit bundles should be supported for major review events such as policy promotions, recommendation suppressions, and post-incident analysis.

13. Context Pane Behavior

The right context pane should be reusable across modules. When the user selects an alert, run, recommendation, or incident, the pane should display summary metadata, linked objects, recommended actions, and operator notes.

The pane should not become a dumping ground. It should answer what this object is, why it matters, what it is linked to, and what can be done next.

Selected object type	Pane sections
Alert	Summary, severity, affected object, recommended next step, acknowledge action
Recommendation	Status, confidence, top changes, linked runs, approve/suppress actions
Incident	Severity, owner, timeline, linked objects, resolve controls
Backtest run	Status, top metrics, artifacts, promotion readiness
Paper drift event	Magnitude, benchmark context, linked rebalance, notes

14. Workflow Specifications

The following workflows should be explicitly supported and testable end to end.

Workflow	Happy path	Key exception path
Approve publication	Open staged recommendation -> review warnings -> approve -> audit emitted	Critical incident detected -> suppress instead
Suppress publication	Open queue item -> choose reason -> link incident -> confirm suppression	Reason missing -> block action
Investigate source issue	Open source alert -> inspect affected contexts -> create incident -> decide rerun or suppression	Source recovers but stale recommendation remains -> require refresh review
Review backtest promotion	Open scorecard -> compare	Missing replay support -> fail

Workflow	Happy path	Key exception path
	thresholds -> approve bundle promotion	promotion
Handle paper drift	Open paper discrepancy -> inspect linked recommendation -> note operational assumption -> assign action	Repeated unexplained drift -> escalate incident

15. States, Notifications, and Empty Cases

Admin UX should treat state handling explicitly. Required states include healthy, warning, degraded, blocked, empty-but-healthy, loading, stale, and action-completed. Empty states are especially important for queues such as no active incidents or no pending approvals; they should reassure without looking broken.

16. Permissions and Safety

- Read-only users should never see destructive or publication-changing actions.
- Approve, suppress, promote, and rollback actions require elevated permissions.
- High-risk actions should require confirmation and should capture a reason or note.
- Context-preserving deep links should respect permissions and hide unavailable actions gracefully.
- Operator notes should be editable only by permitted roles and preserved historically where relevant.

17. Acceptance Criteria

1. Each module answers a distinct operator question without requiring raw log access.
2. Publication, suppression, promotion, and incident actions emit audit events.
3. Queues, banners, and panes use consistent status semantics with the user-facing product.
4. Linked navigation between alert, recommendation, replay, and incident works without losing context.
5. Critical workflows are testable by fixtures and smoke checks.
6. The workspace remains usable on desktop and credible on tablet-sized displays.

Appendix A - Screen Inventory

Suggested named screens: Admin Home, Publication Queue, Recommendation Review, Source Control, Engine Health, Incident Board, Incident Detail, Replay Investigator, Backtest Queue, Backtest Detail, Paper Monitor, Policy Catalog, Audit Explorer, Access Settings.

Appendix B - Open UX Questions

- Whether approval and suppression should live on one combined queue or two filtered queue modes.
- Whether operator notes should support structured tags for postmortem reuse.
- Whether mobile admin access should be view-only in the first release.
- Whether a dedicated compare mode is needed for policy bundle promotions.